

Marketing Mix Modeling using Google Meridian

Abstract

Marketing is one of the biggest investments made by businesses, but understanding which marketing activities actually influence customer purchases can be challenging. This project applies Google's Meridian Marketing Mix Modeling framework to evaluate the effectiveness of different digital marketing channels. Using historical marketing spend and purchase data, the model estimates the contribution of each channel, measures return on investment (ROI), and identifies how advertising effectiveness changes as spending increases. SHAP values are also used to improve model interpretability, while budget optimization explores different spending scenarios to identify more efficient marketing strategies. The results provide practical insights into how marketing budgets can be allocated more effectively to improve business performance.

1. Introduction

Companies invest significant amounts of money in digital marketing through channels such as Google Search, Shopping, Performance Max, and Display campaigns. While these campaigns generate large volumes of data, it is often difficult to determine how much each channel contributes to customer purchases. Understanding this relationship is essential for making informed marketing decisions and ensuring that advertising budgets are spent efficiently.

Marketing Mix Modeling (MMM) is a statistical approach that estimates the contribution of different marketing activities while accounting for external factors and the delayed effects of advertising. In this project, Google's Meridian framework is used to analyze historical marketing data and evaluate the effectiveness of several digital advertising channels.

The main objectives of this project are:

- To measure the effectiveness of different marketing channels.
- To evaluate return on investment and marginal return on investment.
- To explain model predictions using SHAP values.
- To optimize marketing budget allocation under different investment scenarios.

2. Dataset Description

The dataset used in this project contains historical marketing and purchase information collected across multiple organizations and time periods. It includes daily advertising expenditure, customer purchases, click data, and geographical information.

The target variable for the analysis is **ALL_PURCHASES**, representing the total number of purchases made during each time period. Marketing expenditure from four Google advertising channels was selected for modeling:

- Google Paid Search
- Google Shopping
- Google Performance Max
- Google Display

Control variables, including **Year** and **Month**, were also included to capture seasonal trends that may influence customer purchasing behaviour.

Since Google's Meridian framework requires consistent observations across all geographical units, the organization with the longest complete historical record was selected for the final analysis.

3. Data Preparation

Before building the model, several preprocessing steps were carried out to improve data quality and ensure compatibility with the Meridian framework.

Duplicate records were identified and combined by aggregating marketing variables while preserving the corresponding date and geographical information. Date values were converted into an appropriate format, and additional variables representing the year and month were created to capture seasonal effects.

A population variable, required by Meridian, was also added to the dataset. Finally, the organization with the most complete historical data was selected to satisfy the balanced time-series requirement of the modeling framework.

These preprocessing steps resulted in a clean and consistent dataset that was suitable for marketing mix modeling.

4. Model Development

Google Meridian was used to build the marketing mix model. The prepared dataset was converted into Meridian's required format by specifying the target variable, media channels, control variables, and population information.

The model estimates the relationship between advertising expenditure and customer purchases using Bayesian inference. This approach allows uncertainty to be incorporated into the analysis while providing robust estimates of marketing effectiveness.

After the model was configured, prior and posterior sampling procedures were performed to estimate the final model parameters.

5. Model Validation

Once the model was trained, its performance was evaluated using several validation metrics.

The model achieved a health score of approximately **81.3**, indicating stable convergence during Bayesian sampling. In addition, the model produced an R^2 value of **0.4646**, suggesting that it explains a meaningful proportion of the variation in customer purchases. The obtained MAPE and weighted MAPE values also indicate an acceptable level of predictive performance for marketing mix analysis.

Overall, the validation results suggest that the model provides a reliable foundation for analysing marketing effectiveness.

6. Marketing Effectiveness Analysis

Several analyses were performed to better understand the contribution of each marketing channel.

Return on Investment (ROI)

ROI measures how efficiently each advertising channel converts marketing spend into customer purchases. Channels with higher ROI generated greater value for every unit of advertising expenditure.

Marginal Return on Investment (mROI)

Unlike traditional ROI, marginal ROI evaluates the expected benefit of investing additional budget into a marketing channel. This metric helps identify channels that still have potential for further investment.

Channel Contribution

The contribution analysis estimates how much each advertising channel contributes to overall purchases. This provides a clearer understanding of which channels play the largest role in driving business performance.

Hill Curve Analysis

Hill curves were used to examine the relationship between advertising spend and marketing response. The results indicate that some channels begin to experience diminishing returns after reaching higher spending levels, suggesting that simply increasing budget does not always produce proportional increases in purchases.

Adstock Analysis

Advertising often continues to influence customer behaviour after a campaign has ended. Adstock analysis captures this carryover effect and illustrates how the impact of marketing investment changes over time.

7. Model Explainability

Although Meridian provides valuable marketing insights, understanding why the model produces certain predictions is equally important.

To improve transparency, SHAP (SHapley Additive exPlanations) values were calculated using a Random Forest model trained on the same marketing variables. The SHAP summary plots identify the features that have the greatest influence on purchase predictions, while the feature importance plots provide a clear ranking of variable importance.

This additional analysis makes the results easier to interpret and supports more informed marketing decisions.

8. Budget Optimization

One of the advantages of Meridian is its ability to recommend improved budget allocations.

Using the trained model, four different budget allocation scenarios were evaluated:

- Increasing the overall marketing budget by 20%.
- Increasing investment in Google Paid Search by 50%.
- Reducing investment in channels showing signs of saturation.
- Redistributing part of the marketing budget across selected channels.

For each scenario, the optimizer estimated the expected purchases, return on investment, and recommended spending allocation. Comparing these scenarios helped identify strategies that could improve marketing performance while making more efficient use of available resources.

9. Business Recommendations

Based on the findings of this analysis, several recommendations can be made.

Marketing investment should continue to focus on channels that consistently deliver strong returns on investment while avoiding unnecessary increases in channels that have already reached saturation. Rather than simply increasing the total marketing budget, reallocating spending toward more effective channels appears to provide better overall performance.

The use of budget optimization can support future campaign planning by allowing marketers to evaluate multiple investment strategies before launching campaigns. In addition, combining Meridian with SHAP explainability helps improve confidence in the model by providing clear explanations of the factors influencing customer purchases.

10. Limitations

This project has several limitations that should be considered when interpreting the results.

The analysis was conducted using data from a single organization because Meridian requires balanced historical observations across all geographical units. As a result, the findings may not fully represent the behaviour of all organizations within the dataset.

In addition, factors such as competitor activities, pricing changes, promotions, and broader economic conditions were not included in the model. Finally, the optimization results assume that historical relationships between marketing expenditure and customer purchases will remain similar in future campaigns.

11. Conclusion

This project demonstrates how Google Meridian can be used to evaluate marketing effectiveness and support data-driven decision-making.

The model successfully estimated the contribution of different marketing channels, measured advertising efficiency, explained prediction behaviour using SHAP values, and explored alternative budget allocation strategies through optimization scenarios. Overall, the findings show that marketing mix modeling can provide valuable guidance for improving advertising investment decisions and developing more effective marketing strategies in the future.